Field	Value
Document No.	QP-006
Title	In-Process Inspection
Revision	B
Effective Date	2026-08-10
Prepared By	Owner
Approved By	Jacob Honick, Owner
Revision Approval Date	2026-11-05
Next Review	2027-08-10

QP-006 In-Process Inspection

Standards and clause references are guidance only and do not claim certification or standard compliance. Applicable customer, contractual and legal requirements govern accepted work.

1. Purpose

Define the process for performing inspections during production operations to verify conformance at critical stages and detect nonconformances before further value is added to the part.

2. Scope

Applies to all in-process inspection activities including first piece inspection, intermediate dimensional checks, and inspection hold points identified on the job traveler.

3. Responsibilities

Role	Responsibility
Shop Owner	Perform all in-process inspections. Record results on inspection records. Initiate nonconformance reports when out-of-tolerance conditions are found. Verify measurement equipment is current on calibration.

4. References

- QF-005 — In-Process Inspection Record
- QF-011 — Nonconforming Material Report
- WI-005 — First Piece Inspection
- QP-005 — Production Planning and Control
- QP-008 — Nonconforming Product Control
- QP-010 — Calibration and Measurement Control
- AS9100 Rev D Clause 8.5.1, 8.6

5. Definitions

- **First Piece Inspection:** Inspection of the first part produced in a setup to verify the operation will produce conforming product before running the remainder of the batch.
- **Inspection Hold Point:** A mandatory stop in the operation sequence where production cannot continue until inspection is performed and signed off.
- **SPC (Statistical Process Control):** Use of measured data to monitor process stability. Applied when batch sizes and customer requirements warrant.

6. Procedure

First Piece Inspection

1. After completing setup for any machining operation, produce the first piece.
2. Perform first piece inspection per WI-005, measuring all characteristics produced in that operation.
3. Record all measurements on QF-005 In-Process Inspection Record.
4. Compare measured values to drawing tolerances:
    - If all characteristics conform: sign off QF-005, release the setup for production run.
    - If any characteristic is out of tolerance: adjust the setup, produce a new first piece, and re-inspect. Do not run production until first piece passes.
5. For CNC operations: verify program offsets match the first piece measurements before continuing.

In-Process Checkpoints

1. During production runs, perform dimensional checks at intervals appropriate to the operation:
    - Default: inspect every part for all characteristics produced at that operation.

    - Use a different inspection extent only when specified by the customer or expressly accepted by the customer in writing; record the requirement and plan on the traveler.
    - After any tool change or offset adjustment: re-verify the next piece before continuing.
2. Record checkpoint measurements on QF-005.

Inspection Hold Points

1. At operations marked as inspection hold points on the job traveler (QF-004):
    - Stop production after completing the operation.
    - Perform full inspection of all characteristics produced up to that point.
    - Record results on QF-005.
    - Sign off the hold point on QF-004 before releasing to the next operation.
2. Do not proceed past a hold point without documented sign-off.

Statistical Process Control

1. When batch sizes exceed 25 pieces and customer or internal requirements specify SPC:
    - Measure the specified characteristic(s) on each piece unless the customer specifies or approves a different plan in writing. SPC does not replace full acceptance inspection by default.
    - Plot data on X-bar/R chart or individual/moving range chart as appropriate.
    - Investigate any out-of-control condition (point beyond control limits, run of 7, trend of 7).
2. SPC data is recorded on QF-005 or an attached data sheet referenced by the QF-005 number.

Nonconformance Discovery

1. When any in-process measurement reveals a nonconforming condition:
    - Immediately stop production on that operation.
    - Segregate the nonconforming part(s) and any suspect parts produced since the last good check.
    - Initiate QF-011 Nonconforming Material Report per QP-008.
    - Do not resume production until the cause is identified and corrected.
2. Record the NCR number on QF-005 for traceability.

Equipment Verification

1. Before performing any in-process inspection, verify that the measurement equipment being used is within its calibration due date per QP-010.

2. If calibration is overdue, do not use the instrument. Select an alternative calibrated instrument or have the equipment calibrated before proceeding.

7. Records Generated

Form No.	Title	When Completed
QF-005	In-Process Inspection Record	At first piece, each checkpoint, and each hold point
QF-011	Nonconforming Material Report	When any in-process nonconformance is discovered

8. Revision History

Rev	Date	Description	Approved By
A	2026-08-07	Initial release	Shop Owner

Revision B Approval

Approved 2026-11-05 by Jacob Honick, Owner. Release approved Rev B; correct references and make full inspection the default unless the customer specifies or approves an alternative.